Assignment 2

Q1: NOT Gate using 2-input NAND

EDA Playground Link

Verilog Code (Screenshot):

testbench.sv

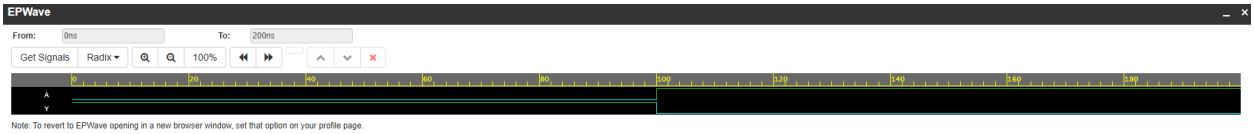
```
module tb_not;
reg A;
wire Y;
initial
begin
$dumpfile ("dump.vcd");
$dumpvars (1, tb_not);
#000 A=0;
#100 A=1;
#100 $stop;
end
not_using_nand U1(A, Y);
endmodule
```

design.sv

```
module nand_gate (A, B, Y);
input A, B;
output Y;
assign Y = ~(A & B);
endmodule

module not_using_nand (A, Y);
input A;
output Y;
nand_gate N1(A, A, Y);
endmodule
```

EDA Playground Waveform (Screenshot):



Q2: Dataflow Logic (Y = ABC + AB + AC)

EDA Playground Link

Verilog Code (Screenshot):

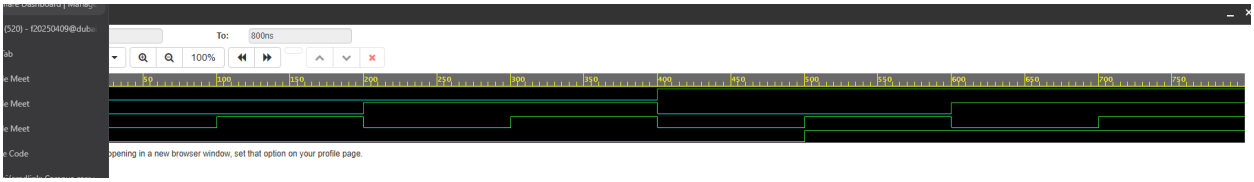
testbench.sv

```
module tb_logic;
reg A, B, C;
wire Y;
initial
begin
$dumpfile ("dump.vcd");
$dumpvars (1, tb_logic);
#000 A=0; B=0; C=0;
#100 A=0; B=0; C=1;
#100 A=0; B=1; C=0;
#100 A=0; B=1; C=1;
#100 A=1; B=0; C=0;
#100 A=1; B=0; C=1;
#100 A=1; B=1; C=0;
#100 A=1; B=1; C=1;
#100 $stop;
end
logic_dataflow U1(A, B, C, Y);
endmodule
```

design.sv

```
module logic_dataflow (A, B, C, Y);
input A, B, C;
output Y;
assign Y = (A & B & C) | (A & B) | (A & C);
endmodule
```

EDA Playground Waveform (Screenshot):



Q3: 4-bit Even Parity Generator

EDA Playground Link

Verilog Code (Screenshot):

testbench.sv

```
module tb_parity;
reg A, B, C, D;
wire P;
initial
begin
$dumpfile ("dump.vcd");
$dumpvars (1, tb_parity);
#000 A=0; B=0; C=0; D=0;
#100 A=0; B=0; C=0; D=1;
#100 A=0; B=0; C=1; D=0;
#100 A=0; B=0; C=1; D=1;
#100 A=0; B=1; C=0; D=0;
#100 A=0; B=1; C=0; D=1;
#100 A=0; B=1; C=1; D=0;
#100 A=0; B=1; C=1; D=1;
#100 A=1; B=0; C=0; D=0;
#100 A=1; B=0; C=0; D=1;
#100 A=1; B=0; C=1; D=0;
#100 A=1; B=0; C=1; D=1;
#100 A=1; B=1; C=0; D=0;
#100 A=1; B=1; C=0; D=1;
#100 A=1; B=1; C=1; D=0;
#100 A=1; B=1; C=1; D=1;
#100 $stop;
end
parity_4bit U1(A, B, C, D, P);
endmodule
```

design.sv

```
module parity_4bit (A, B, C, D, P);
input A, B, C, D;
output P;
assign P = A ^ B ^ C ^ D;
endmodule
```

EDA Playground Waveform (Screenshot):

